

CAPSTONE PROJECT

PHOTOVOLTAIC ENERGY UTILIZATION AND CURTAILMENT ANALYSIS IN A HYBRID SOLAR-DIESEL SYSTEM

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1. ASK PHASE

1.1 Introduction

This case study analyzes the operational behavior of the El Sena hybrid solar-diesel energy system located in Pando, Bolivia. Using operational and environmental data recorded between October 2018 and September 2019, the project explores how electrical demand, diesel generation, and solar radiation affect photovoltaic energy utilization.



Figure 1. *El Sena* Solar Photovoltaic Plant Location in Bolivia

Source: SIE – Sistemas Integrales de Energía

The analysis focuses on identifying patterns of solar energy curtailment and operational limitations that may reduce the effective use of renewable energy within the system.

1.2 Business Task

Analyze operational and environmental data from the El Sena hybrid solar-diesel system to identify patterns in photovoltaic energy utilization, diesel dependence, and solar energy curtailment.

1.3 Problem Statement

Hybrid solar-diesel systems can improve energy sustainability in isolated regions; however, operational constraints may limit the amount of photovoltaic energy that can be injected into the electrical system. As a result, part of the available solar energy may remain unused despite favorable environmental conditions.

This project investigates how solar radiation, electrical demand, and operational limits influence photovoltaic energy utilization in the El Sena Solar Plant. The analysis may help stakeholders better understand periods of solar energy curtailment and explore opportunities to improve renewable energy integration and system efficiency.

1.4 Stakeholders

The main stakeholders for this project include renewable energy organizations, hybrid system operators, energy engineers, researchers, and professionals interested in renewable energy integration and operational efficiency.

1.5 Business Questions

- How does photovoltaic energy utilization vary throughout the day and across different months?
- During which periods is solar energy most underutilized?
- How does solar radiation influence photovoltaic generation potential?
- What relationship exists between electrical demand and photovoltaic energy injection?
- During which periods does the system rely more heavily on diesel generation?
- How much potential photovoltaic energy remains unused due to operational constraints?

1.6 Metrics

Key metrics analyzed in this project include:

- Electrical demand (kW)
- Diesel generation (kW)
- Photovoltaic energy injected (kW)
- Unused photovoltaic energy (kW)
- Solar radiation (kWh/m²)

Additional performance indicators such as photovoltaic utilization patterns, diesel dependence, and estimated curtailment periods will also be explored during the analysis.

1.7 Dataset Overview

The dataset contains operational data from the El Sena Solar Photovoltaic Plant, a hybrid solar-diesel system located in Pando, Bolivia. Measurements were recorded every 15 minutes for selected operational days between October 2018 and September 2019.

The dataset includes variables related to electrical demand, diesel generator output, solar radiation, photovoltaic generation potential, photovoltaic injection limits, and unused photovoltaic energy. The system also contains operational constraints that determine when and how much photovoltaic energy can be injected into the electrical network.

1.8 Expected Outcomes

This analysis is expected to identify patterns in photovoltaic energy utilization, quantify periods of unused solar energy, evaluate diesel generation dependence, and explore operational limitations affecting renewable energy integration.

The project also aims to demonstrate the application of data analytics techniques in hybrid renewable energy systems and generate insights related to potential energy storage opportunities.

2. PREPARE PHASE

2.1 Data Source

The dataset used in this project was obtained from operational records of the El Sena Solar Photovoltaic Plant, a hybrid solar-diesel energy system located in the department of Pando, Bolivia.

The data was provided through a personal collaboration and contains historical operational records from selected days between October 2018 and September 2019.

2.2 Data Organization

The dataset was originally organized into 12 separate monthly tables corresponding to selected operational days from different months between 2018 and 2019. Each table contains measurements recorded at 15-minute intervals over a 24-hour period.

Hora	Dec-18			Potencia Mínima que puede ingresar	Ingresa o no	Cuanto potencia fotovoltaica ingresara (Potencial de inyección)	Generación Fotovoltaica Teórica (Kw)	Cuanta fotovoltaica puede entrar (Kw)	Cuanta fotovoltaica queda (Kw)
	Demanda (Kw)	Generacion Nominal Diesel (Kw)	Radiacion (Kw*h/m^2)	52%					
0:00	300.26	700	0	364	0	0	0	0	0
0:15	293.46	700	0	364	0	0	0	0	0
0:30	288.28	700	0	364	0	0	0	0	0
0:45	283.7	700	0	364	0	0	0	0	0
1:00	279.4	700	0	364	0	0	0	0	0
1:15	275.86	700	0	364	0	0	0	0	0
1:30	270.88	700	0	364	0	0	0	0	0
1:45	267.63	700	0	364	0	0	0	0	0
2:00	265.5	300	0	156	1	109.5	0	0	0
2:15	262.64	300	0	156	1	106.64	0	0	0
2:30	261.03	300	0	156	1	105.03	0	0	0
2:45	261.33	300	0	156	1	105.33	0	0	0
3:00	263.2	300	0	156	1	107.2	0	0	0

Figure 2. Sample of the Dataset Used in This Project

The variables include electrical demand, diesel generator power, solar radiation, photovoltaic generation potential, injection constraints, actual photovoltaic injection, and unused photovoltaic energy.

2.3 Data Credibility and ROCCC

The dataset is considered reliable for exploratory analysis because it originates from operational records associated with a real hybrid energy system in Bolivia. Although the data corresponds to the 2018–2019 period and includes some calculated variables, it remains relevant for evaluating operational behavior, photovoltaic utilization, and renewable energy integration patterns.

2.4 Data Integrity

Data integrity was evaluated through manual inspection of the records, verification of time intervals, identification of missing values, and validation of numerical consistency between operational variables.

2.5 Data Privacy and Accessibility

The dataset does not contain personal or sensitive information. The analysis focuses exclusively on operational and environmental energy data.

The dataset used in this project was obtained through academic collaboration and is not publicly distributed. Only analytical results and selected sample records are presented in this portfolio project.

2.6 Potential Data Limitations

Several limitations should be considered during the analysis. The dataset contains selected operational days rather than continuous daily records, and some variables may represent theoretical or calculated values instead of direct measurements.

Additionally, the dataset represents a specific hybrid energy system and may not necessarily generalize to other operational contexts.

3. PROCESS PHASE

3.1 Tools Used

The following tools were used during the data processing phase:

- Microsoft Excel for data consolidation, inspection, cleaning, and formatting
- Google BigQuery for cloud-based SQL processing and future analytical queries

Excel was selected for the initial preparation stage because it allowed efficient manual inspection and consolidation of the monthly operational tables. Google BigQuery was selected to support scalable SQL analysis and to practice cloud-based data analytics workflows.

3.2 Data Consolidation

The original dataset was provided as 12 separate monthly tables corresponding to selected operational days between October 2018 and September 2019.

To facilitate analysis, all tables were consolidated into a single master dataset. During this process, column names were standardized to ensure consistency across all monthly records. Variable names were reformatted using a uniform naming structure to improve readability and compatibility with SQL-based analysis.

The consolidation process was initially performed manually in Microsoft Excel by combining all monthly operational tables into a single structured dataset. During this stage, only calculated values were preserved while original Excel formulas were removed to avoid inconsistencies during cloud-based processing.

Additionally, time formats and numerical structures were standardized before exporting the final dataset to CSV format for integration into Google BigQuery. A new period column was also created to identify the corresponding month and year of each observation.

The final consolidated dataset contained 1,152 operational records measured at 15-minute intervals.

3.3 Data Cleaning Process

The dataset was manually reviewed to identify missing values, formatting inconsistencies, anomalous observations, and numerical inconsistencies.

One blank cell was identified during the validation process and corrected after verifying that the corresponding operational value should be equal to zero.

Additionally, several preprocessing tasks were performed before importing the dataset into Google BigQuery, including:

- Standardization of column names and variable structures
- Reformatting of time values into a consistent format
- Verification of numerical consistency between operational variables
- Removal of unnecessary Excel formulas while preserving calculated values
- Validation of data types for compatibility with SQL-based analysis.

After the CSV file was imported into Google BigQuery, additional SQL validation queries were used to verify dataset integrity, confirm the absence of null values in key operational variables, and validate the total number of imported records.

3.4 Data Transformation

After the cleaning process was completed, the consolidated dataset was exported as a UTF-8 CSV file to ensure compatibility with cloud-based analytical tools.

The CSV file was then imported into Google BigQuery by creating a dedicated Google Cloud project, a BigQuery dataset, and a relational table structure for future SQL-based analysis.

3.5 Data Integrity Verification

Data integrity was verified through manual inspection of records, validation of time interval continuity, verification of numerical consistency, and confirmation of successful CSV import into BigQuery.

Special attention was given to consistency between electrical demand, diesel generation, photovoltaic generation potential, and photovoltaic injection constraints.

The dataset was determined to be sufficiently clean and structured for exploratory analysis.

4. ANALIZE PHASE

4.1 Initial Data Exploration

Initial exploratory analysis was performed in Google BigQuery to better understand the structure, consistency, and operational behavior of the dataset before conducting deeper analysis.

The first validation queries focused on:

- confirming the total number of records,
- verifying the temporal coverage of the dataset,
- checking for missing values,
- and generating descriptive statistics for the main operational variables.

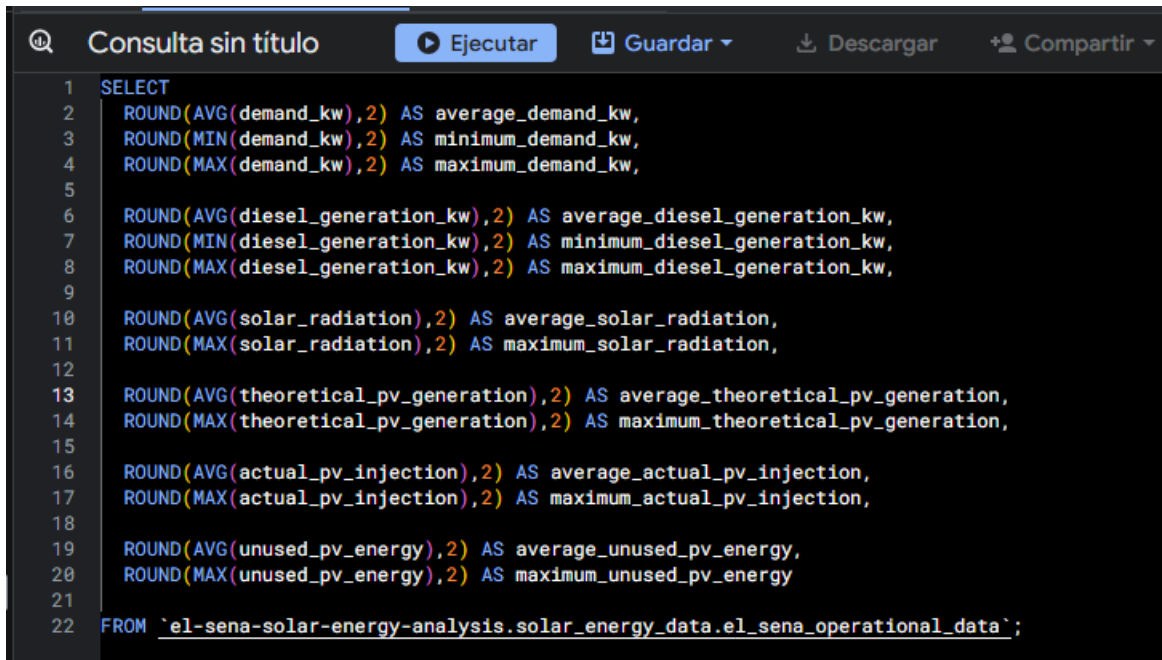
The consolidated dataset contained 1,152 operational records collected at 15-minute intervals across 12 selected operational days between October 2018 and September 2019.

SQL validation confirmed that no null values remained in the key operational variables after the data cleaning process, supporting the reliability and consistency of the dataset for exploratory analysis.

Descriptive statistics were then generated to better understand the operational ranges and overall behavior of the hybrid solar-diesel system.

Variable	Average	Minimum	Maximum
Electrical Demand (kW)	298.61	164.33	468.12
Diesel Generation (kW)	552.17	300.00	1100.00

Solar Radiation	197.59	-	803.97
Theoretical PV Generation (kW)	84.24	-	342.78
Actual PV Injection (kW)	22.34	-	180.97
Unused PV Energy (kW)	61.91	-	330.33



```

1 SELECT
2   ROUND(AVG(demand_kw),2) AS average_demand_kw,
3   ROUND(MIN(demand_kw),2) AS minimum_demand_kw,
4   ROUND(MAX(demand_kw),2) AS maximum_demand_kw,
5
6   ROUND(AVG(diesel_generation_kw),2) AS average_diesel_generation_kw,
7   ROUND(MIN(diesel_generation_kw),2) AS minimum_diesel_generation_kw,
8   ROUND(MAX(diesel_generation_kw),2) AS maximum_diesel_generation_kw,
9
10  ROUND(AVG(solar_radiation),2) AS average_solar_radiation,
11  ROUND(MAX(solar_radiation),2) AS maximum_solar_radiation,
12
13  ROUND(AVG(theoretical_pv_generation),2) AS average_theoretical_pv_generation,
14  ROUND(MAX(theoretical_pv_generation),2) AS maximum_theoretical_pv_generation,
15
16  ROUND(AVG(actual_pv_injection),2) AS average_actual_pv_injection,
17  ROUND(MAX(actual_pv_injection),2) AS maximum_actual_pv_injection,
18
19  ROUND(AVG(UNUSED_PV_ENERGY),2) AS average_unused_pv_energy,
20  ROUND(MAX(UNUSED_PV_ENERGY),2) AS maximum_unused_pv_energy
21
22 FROM `el-sena-solar-energy-analysis.solar_energy_data.el_sena_operational_data`;

```

The descriptive analysis revealed that diesel generation remained relatively high throughout the analyzed operational periods, while actual photovoltaic injection remained considerably lower than the theoretical photovoltaic generation potential.

Additionally, the dataset showed substantial levels of unused photovoltaic energy, suggesting that operational limitations may have restricted the full utilization of available solar energy within the hybrid system.

4.2 Hourly Operational Analysis

An hourly operational analysis was performed to evaluate daily variations in:

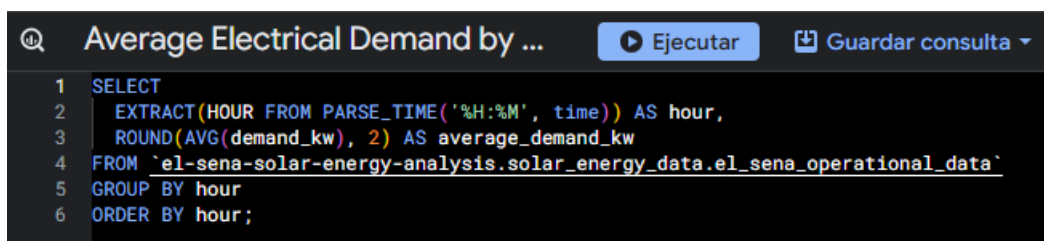
- Electrical demand,
- Diesel generation,
- Solar radiation,
- Actual photovoltaic injection,
- Unused photovoltaic energy.

The analysis grouped operational records by hour and calculated average values for each operational variable.

4.2.1 Electrical Demand

The results revealed clear daily electricity consumption patterns throughout the operational periods analyzed.

Hour	Avg Demand (kW)	Hour	Avg Demand (kW)	Hour	Avg Demand (kW)
0	242.64	8	279.38	16	355.60
1	226.38	9	310.12	17	343.78
2	217.60	10	329.43	18	355.26
3	212.73	11	344.33	19	401.16
4	209.83	12	324.41	20	392.85
5	213.05	13	327.08	21	363.52
6	201.46	14	345.18	22	315.11
7	224.27	15	359.67	23	271.76



```

1 SELECT
2   EXTRACT(HOUR FROM PARSE_TIME('%H:%M', time)) AS hour,
3   ROUND(AVG(demand_kw), 2) AS average_demand_kw
4 FROM `el-sena-solar-energy-analysis.solar_energy_data.el_sena_operational_data`
5 GROUP BY hour
6 ORDER BY hour;

```

Electrical demand remained relatively low during early morning hours and gradually increased throughout daytime and evening operation. The highest average demand levels were observed between 19:00 and 20:00, when average demand exceeded 390 kW.

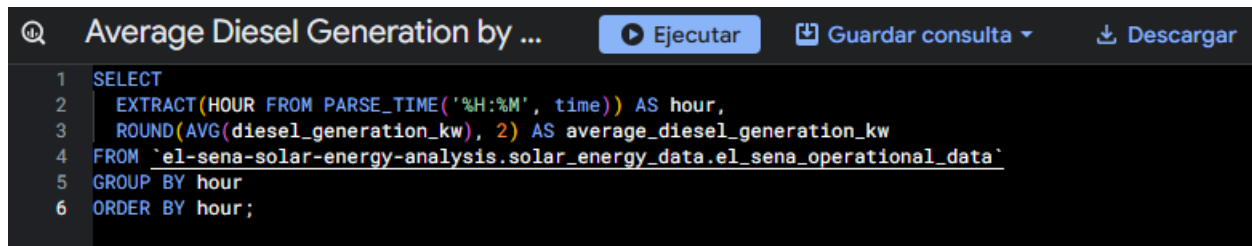
This behavior suggests that peak electricity consumption occurred during nighttime operational periods, while lower demand levels were recorded during early morning hours.

4.2.2 Diesel Generation

Diesel generation remained relatively high throughout most operational periods.

Lower diesel generation levels were observed during early morning hours, while generation gradually increased during daytime and evening operation. The highest diesel generation values were recorded between 19:00 and 21:00, reaching approximately 800 kW during peak demand periods.

Hour	Avg Diesel Generation (kW)	Hour	Avg Diesel Generation (kW)	Hour	Avg Diesel Generation (kW)
0	508.33	8	487.50	16	722.92
1	341.67	9	475.00	17	758.33
2	306.25	10	475.00	18	772.92
3	300.00	11	535.42	19	800.00
4	302.08	12	512.50	20	800.00
5	306.25	13	568.75	21	791.67
6	393.75	14	581.25	22	716.67
7	450.00	15	645.83	23	700.00



```

1 SELECT
2   EXTRACT(HOUR FROM PARSE_TIME('%H:%M', time)) AS hour,
3   ROUND(AVG(diesel_generation_kw), 2) AS average_diesel_generation_kw
4 FROM `el-sena-solar-energy-analysis.solar_energy_data.el_sena_operational_data`
5 GROUP BY hour
6 ORDER BY hour;

```

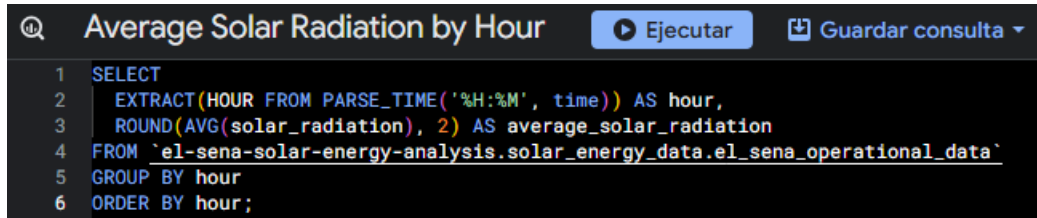
Additionally, diesel generation remained considerable even during daytime hours when photovoltaic generation potential was available. This operational behavior may suggest continued dependence on diesel generation to maintain system stability and supply reliability.

4.2.3 Solar Radiation

The solar radiation analysis showed a clear daytime solar resource pattern consistent with expected photovoltaic operating conditions.

Hour	Avg Solar Radiation	Hour	Avg Solar Radiation	Hour	Avg Solar Radiation
0	0.00	8	285.20	16	263.89
1	0.00	9	443.52	17	111.41
2	0.00	10	578.42	18	11.74
3	0.00	11	658.79	19	0.01
4	0.00	12	682.05	20	0.00

5	0.09	13	639.17	21	0.00
6	15.61	14	527.52	22	0.00
7	117.88	15	406.81	23	0.00



```

1 SELECT
2   EXTRACT(HOUR FROM PARSE_TIME('%H:%M', time)) AS hour,
3   ROUND(AVG(solar_radiation), 2) AS average_solar_radiation
4 FROM `el-sena-solar-energy-analysis.solar_energy_data.el_sena_operational_data`
5 GROUP BY hour
6 ORDER BY hour;

```

Solar radiation remained near zero during nighttime hours, increased after sunrise, and reached its highest average levels between approximately 11:00 and 13:00. Peak average solar radiation reached approximately 682 during midday periods.

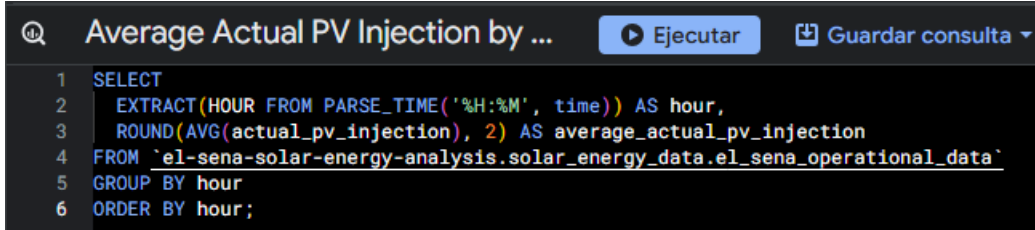
These results confirmed the presence of significant solar resource availability during daytime operation, creating favorable conditions for photovoltaic generation within the hybrid system.

4.2.4 Actual Photovoltaic Injection

Actual photovoltaic injection followed the general daytime solar radiation pattern, increasing after sunrise and gradually decreasing during afternoon and evening hours.

The highest average photovoltaic injection value was recorded at 10:00, reaching approximately 89.18 kW.

Hour	Avg Actual PV Injection (kW)	Hour	Avg Actual PV Injection (kW)	Hour	Avg Actual PV Injection (kW)
0	0.00	8	46.28	16	19.19
1	0.00	9	75.27	17	6.48
2	0.00	10	89.18	18	1.22
3	0.00	11	77.49	19	0.00
4	0.00	12	68.11	20	0.00
5	0.04	13	43.84	21	0.00
6	1.91	14	50.01	22	0.00
7	17.95	15	39.08	23	0.00



```

1 SELECT
2   EXTRACT(HOUR FROM PARSE_TIME('%H:%M', time)) AS hour,
3   ROUND(AVG(actual_pv_injection), 2) AS average_actual_pv_injection
4 FROM `el-sena-solar-energy-analysis.solar_energy_data.el_sena_operational_data`
5 GROUP BY hour
6 ORDER BY hour;

```

However, despite the presence of significant solar radiation during midday periods, actual photovoltaic injection remained relatively limited when compared with the theoretical photovoltaic generation potential identified in the descriptive analysis. This behavior may indicate the presence of operational constraints affecting renewable energy integration within the system.

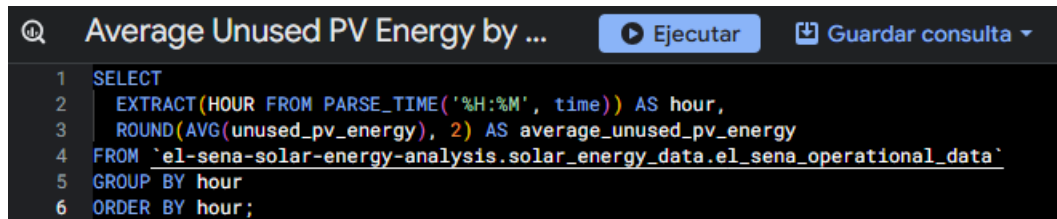
4.2.5. Unused PV Energy

The analysis revealed considerable levels of unused photovoltaic energy during daytime operational periods.

Unused photovoltaic energy increased significantly after sunrise and reached its highest average values between approximately 11:00 and 14:00, coinciding with periods of strongest solar radiation and highest theoretical photovoltaic generation potential.

The maximum average unused photovoltaic energy value was recorded at 13:00, reaching approximately 228.67 kW.

Hour	Avg Unused PV Energy (kW)	Hour	Avg Unused PV Energy (kW)	Hour	Avg Unused PV Energy (kW)
0	0.00	8	75.32	16	93.32
1	0.00	9	113.84	17	41.02
2	0.00	10	157.43	18	3.79
3	0.00	11	203.39	19	0.00
4	0.00	12	222.69	20	0.00
5	0.00	13	228.67	21	0.00
6	4.74	14	174.91	22	0.00
7	32.31	15	134.37	23	0.00



```

1 SELECT
2   EXTRACT(HOUR FROM PARSE_TIME('%H:%M', time)) AS hour,
3   ROUND(AVG(UNUSED_PV_ENERGY), 2) AS average_unused_pv_energy
4 FROM `el-sena-solar-energy-analysis.solar_energy_data.el_sena_operational_data`
5 GROUP BY hour
6 ORDER BY hour;

```

These results suggest that a substantial portion of the available photovoltaic generation potential could not be fully utilized during peak solar production hours. The persistence of high unused photovoltaic energy levels during periods of strong solar resource availability reinforces evidence of photovoltaic curtailment within the hybrid solar-diesel system.

Additionally, this behavior may support the evaluation of future energy storage solutions or operational optimization strategies to improve renewable energy utilization.

4.3 Monthly Operational Analysis

A monthly operational analysis was performed to evaluate variations in:

- Photovoltaic utilization,
- Diesel dependence,
- Electrical demand,
- Solar radiation,
- Photovoltaic curtailment across the analyzed operational periods.

4.3.1 Monthly Unused Photovoltaic Energy

The analysis revealed noticeable variations in unused photovoltaic energy across different operational periods.

Higher average unused photovoltaic energy levels were observed during October, November, December, August, and September, while lower average values were recorded between January and March.

The highest average unused photovoltaic energy value was observed in December 2018, reaching approximately 81.24 kW.

Period	Avg Unused PV Energy (kW)	Period	Avg Unused PV Energy (kW)
2018-10	61.59	2019-04	59.37
2018-11	77.43	2019-05	53.34
2018-12	81.24	2019-06	56.24
2019-01	48.24	2019-07	66.12

2019-02	46.12	2019-08	69.26
2019-03	44.58	2019-09	79.36

```

Average Unused PV Energy by ...
Ejecutar Guardar consulta
1 SELECT
2   period,
3   ROUND(AVG(unused_pv_energy), 2) AS average_unused_pv_energy
4 FROM `el-sena-solar-energy-analysis.solar_energy_data.el_sena_operational_data`
5 GROUP BY period
6 ORDER BY period;

```

These results suggest that photovoltaic curtailment was not constant throughout the analyzed operational periods and may have been influenced by changing operational conditions, solar resource availability, or system demand behavior.

4.3.2 Monthly Actual Photovoltaic Injection

Significant variations in photovoltaic utilization were observed between operational periods.

The highest average photovoltaic injection values were recorded during March and May 2019, when average photovoltaic injection exceeded 34 kW. In contrast, photovoltaic injection remained extremely low during November and December 2018, with average values remaining close to zero.

Period	Avg Actual PV Injection (kW)	Period	Avg Actual PV Injection (kW)
2018-10	30.57	2019-04	29.72
2018-11	0.39	2019-05	34.51
2018-12	0.12	2019-06	31.17
2019-01	24.05	2019-07	20.81
2019-02	26.31	2019-08	25.18
2019-03	34.34	2019-09	10.84

```

Average Actual PV Injection by ...
Ejecutar Guardar consulta
1 SELECT
2   period,
3   ROUND(AVG(actual_pv_injection), 2) AS average_actual_pv_injection
4 FROM `el-sena-solar-energy-analysis.solar_energy_data.el_sena_operational_data`
5 GROUP BY period
6 ORDER BY period;

```


These results are particularly important because they suggest that photovoltaic generation was heavily limited during certain operational periods despite the presence of available solar resource conditions.

A particularly notable operational anomaly was observed during November and December 2018, when actual photovoltaic injection remained close to zero even though solar radiation levels remained within normal operational ranges. Since electrical demand during these periods was not exceptionally high, the results suggest that factors other than solar resource availability or demand behavior may have affected photovoltaic integration.

One possible business-oriented hypothesis is that the system may have experienced operational restrictions such as inverter maintenance, dispatch limitations, equipment malfunction, protective operational constraints, or temporary operational strategies prioritizing diesel generation to maintain system stability and reliability.

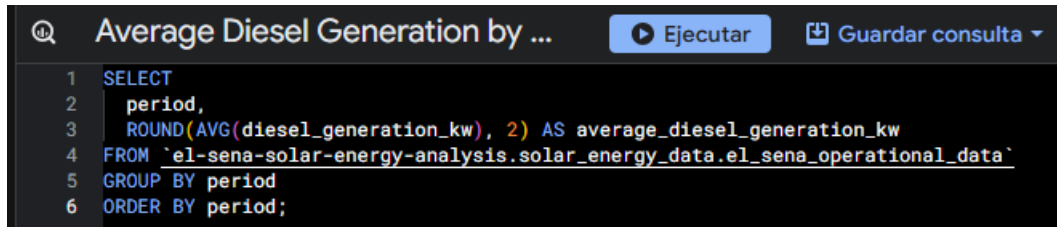
Although the dataset does not include maintenance or operational event logs to confirm these conditions, identifying these anomalies may help stakeholders prioritize future operational investigations and improve renewable energy integration performance.

4.3.3 Monthly Diesel Generation

Diesel generation remained consistently high throughout all analyzed operational periods.

Higher average diesel generation levels were observed during November and December 2018, as well as between July and September 2019. The highest average diesel generation value was recorded in September 2019, reaching approximately 714.58 kW.

Period	Avg Diesel Generation (kW)	Period	Avg Diesel Generation (kW)
2018-10	585.42	2019-04	492.71
2018-11	612.50	2019-05	489.58
2018-12	589.58	2019-06	525.00
2019-01	443.75	2019-07	631.25
2019-02	477.08	2019-08	596.88
2019-03	467.71	2019-09	714.58



```

1 SELECT
2   period,
3   ROUND(AVG(diesel_generation_kw), 2) AS average_diesel_generation_kw
4 FROM `el-sena-solar-energy-analysis.solar_energy_data.el_sena_operational_data`
5 GROUP BY period
6 ORDER BY period;

```

These findings become particularly relevant when compared with photovoltaic injection behavior. During November and December 2018, actual photovoltaic injection remained extremely low while diesel generation continued operating at relatively high levels.

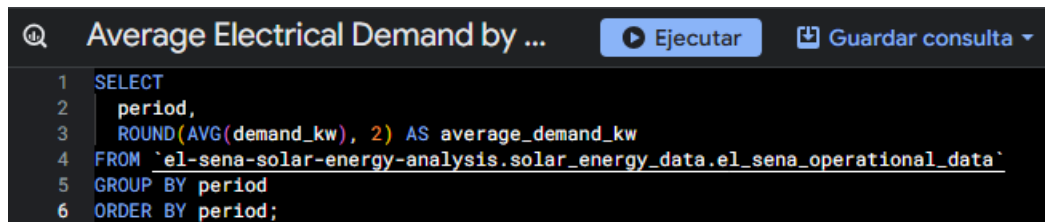
This operational behavior may indicate that the hybrid system relied heavily on diesel generation during periods when photovoltaic integration was limited.

4.3.4 Monthly Electrical Demand

Moderate variations in electrical demand were observed across the analyzed operational periods.

Lower average demand values were recorded during December 2018 and January 2019, while higher average demand levels occurred during October 2018 and September 2019.

Period	Avg Electrical Demand (kW)	Period	Avg Electrical Demand (kW)
2018-10	326.09	2019-04	288.15
2018-11	304.58	2019-05	296.98
2018-12	263.83	2019-06	319.95
2019-01	251.65	2019-07	317.42
2019-02	269.32	2019-08	317.43
2019-03	277.03	2019-09	350.86



```

1 SELECT
2   period,
3   ROUND(AVG(demand_kw), 2) AS average_demand_kw
4 FROM `el-sena-solar-energy-analysis.solar_energy_data.el_sena_operational_data`
5 GROUP BY period
6 ORDER BY period;

```

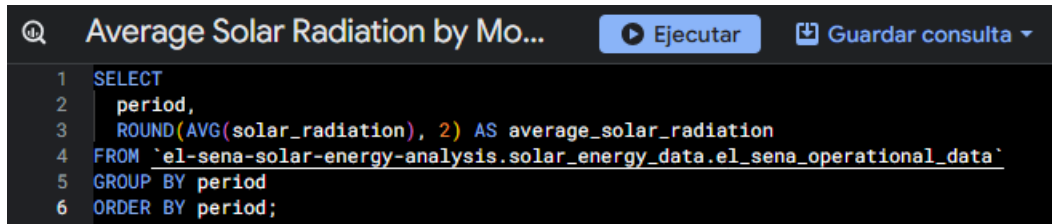
When compared with diesel generation and photovoltaic injection behavior, the results suggest that the decrease in photovoltaic injection observed during November and December 2018 cannot be explained solely by unusually high electrical demand conditions.

4.3.5 Monthly Solar Radiation

Solar radiation levels remained relatively consistent across the analyzed operational periods, with moderate variations between months.

Higher average solar radiation values were observed during October 2018, August 2019, and September 2019, while lower average values were recorded during January and February 2019.

Period	Avg Solar Radiation	Period	Avg Solar Radiation
2018-10	216.18	2019-04	208.96
2018-11	182.52	2019-05	206.04
2018-12	190.83	2019-06	205.02
2019-01	169.57	2019-07	203.91
2019-02	169.87	2019-08	221.49
2019-03	185.10	2019-09	211.57



```

1 SELECT
2   period,
3   ROUND(AVG(solar_radiation), 2) AS average_solar_radiation
4 FROM `el-sena-solar-energy-analysis.solar_energy_data.el_sena_operational_data`
5 GROUP BY period
6 ORDER BY period;

```

This behavior suggests that the low photovoltaic utilization observed during those operational periods may not be explained solely by reduced solar resource availability and may instead indicate the presence of operational constraints affecting photovoltaic energy integration within the hybrid system.

Because solar radiation levels remained relatively stable during November and December 2018 while photovoltaic injection dropped to nearly zero, the results may indicate the occurrence of operational anomalies or temporary technical restrictions within the hybrid system. Possible explanations could include inverter maintenance, equipment malfunction, dispatch prioritization strategies, or operational safety constraints limiting photovoltaic integration.

Although these hypotheses cannot be directly confirmed with the available dataset, incorporating operational maintenance records and equipment event logs in future studies could help explain the causes of these abnormal photovoltaic utilization patterns.

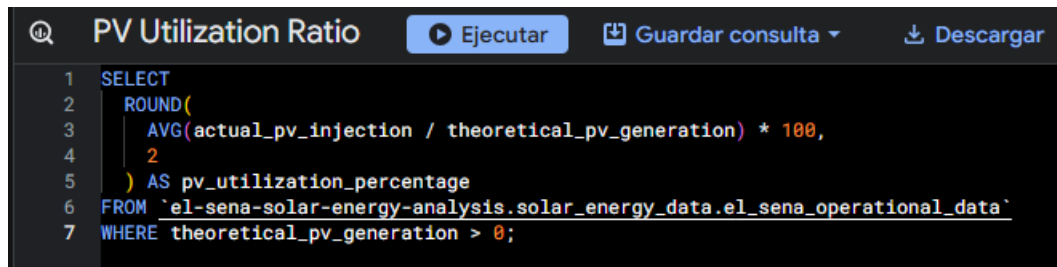
4.4 Photovoltaic Utilization and Curtailment Indicators

To further evaluate renewable energy utilization within the hybrid system, additional photovoltaic performance indicators were calculated.

4.4.1 Photovoltaic Utilization Ratio

The photovoltaic utilization ratio compared the amount of photovoltaic energy actually injected into the system against the theoretical photovoltaic generation potential during periods with available solar generation.

$$\frac{\text{actual_pv_injection}}{\text{theoretical_pv_generation}} \times 100$$



```

1 SELECT
2   ROUND(
3     AVG(actual_pv_injection / theoretical_pv_generation) * 100,
4     2
5   ) AS pv_utilization_percentage
6 FROM `el-sena-solar-energy-analysis.solar_energy_data.el_sena_operational_data`
7 WHERE theoretical_pv_generation > 0;

```

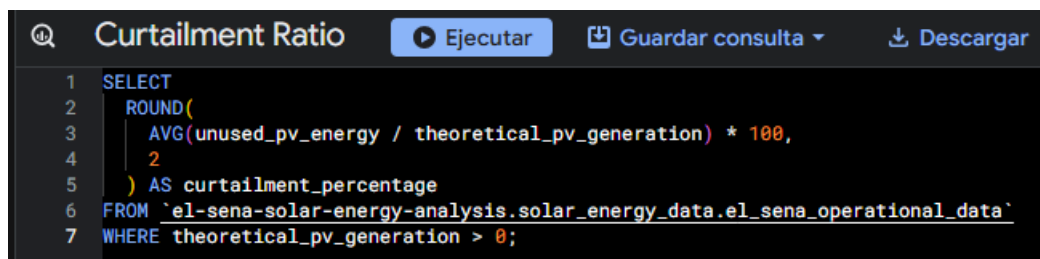
The results showed that, on average, only approximately 26.8% of the theoretical photovoltaic generation potential was effectively injected into the electrical system during periods with available solar generation.

This finding suggests that a substantial portion of the available photovoltaic energy could not be fully utilized, reinforcing evidence of photovoltaic curtailment and operational limitations affecting renewable energy integration within the hybrid solar-diesel system.

4.4.2 Photovoltaic Curtailment Ratio

To quantify the level of unused photovoltaic generation, a photovoltaic curtailment ratio was calculated.

$$\frac{\text{unused_pv_energy}}{\text{theoretical_pv_generation}} \times 100$$



```

1 SELECT
2   ROUND(
3     AVG(unused_pv_energy / theoretical_pv_generation) * 100,
4     2
5   ) AS curtailment_percentage
6 FROM `el-sena-solar-energy-analysis.solar_energy_data.el_sena_operational_data`
7 WHERE theoretical_pv_generation > 0;

```

The results showed that approximately 73.2% of the theoretical photovoltaic generation potential remained unused during periods with available solar generation.

This finding indicates that a substantial portion of the available photovoltaic energy could not be effectively integrated into the electrical system.

4.4.3 Comparative Photovoltaic Energy Summary

A comparative review of photovoltaic generation variables confirmed that average actual photovoltaic injection remained significantly lower than the theoretical photovoltaic generation potential.

Metric	Average Value (kW)
Theoretical PV Generation	84.24
Actual PV Injection	22.34
Unused PV Energy	61.91

The comparison reinforced previous findings indicating substantial levels of unused photovoltaic energy and limited photovoltaic utilization within the hybrid solar-diesel system.

Overall, the analysis suggests that operational limitations significantly affected photovoltaic energy integration during the analyzed operational periods, resulting in considerable levels of photovoltaic curtailment and continued dependence on diesel generation.

5. SHARE PHASE

5.1 Purpose of the Share Phase

The purpose of this phase is to communicate the main findings of the analysis through clear and effective visualizations. Since the project focuses on photovoltaic utilization and curtailment in a hybrid solar-diesel system, the visual presentation should help stakeholders quickly understand how solar resource availability, electrical demand, diesel generation, and operational constraints affect renewable energy integration.

The visualizations were designed to highlight the most important operational patterns discovered during the analysis, especially the low photovoltaic utilization, the high curtailment ratio, and the persistent dependence on diesel generation.

5.2 Audience

The main audience for this presentation includes energy system operators, renewable energy stakeholders, engineers, researchers, and decision-makers interested in hybrid power systems. For this reason, the visualizations were designed to be professional, accessible, and easy to interpret while still preserving technical meaning and analytical value.

Special attention was given to visual clarity, axis labeling, color contrast, and chart simplicity to improve readability and communicate the findings effectively.

5.3 Best Way to Share the Findings

The findings were shared through a series of Tableau visualizations supported by concise written explanations. Tableau was selected because it allows the creation of clear and professional visualizations capable of comparing operational variables across hourly and monthly periods.

The visualizations were organized to support the main business questions by presenting:

- Hourly operational behavior,
- Monthly operational trends,
- Photovoltaic utilization patterns,
- Diesel generation dependence,
- Photovoltaic curtailment indicators.

The visualizations were arranged to progressively explain how solar resource availability, electrical demand, and operational limitations influenced photovoltaic energy integration within the hybrid solar-diesel system.

5.4 Supporting Visualizations and Key Findings

5.4.1 Solar Radiation vs Actual PV Injection by Hour

The visualization shows that solar radiation increased significantly during daytime operation and reached its highest levels between approximately 11:00 and 13:00. However, actual photovoltaic injection remained comparatively limited despite favorable solar resource availability.

Although photovoltaic injection generally followed the same daily pattern as solar radiation, the magnitude of photovoltaic energy injected into the system remained considerably lower than the available solar generation potential.

Solar Radiation vs Actual PV Injection by Hour

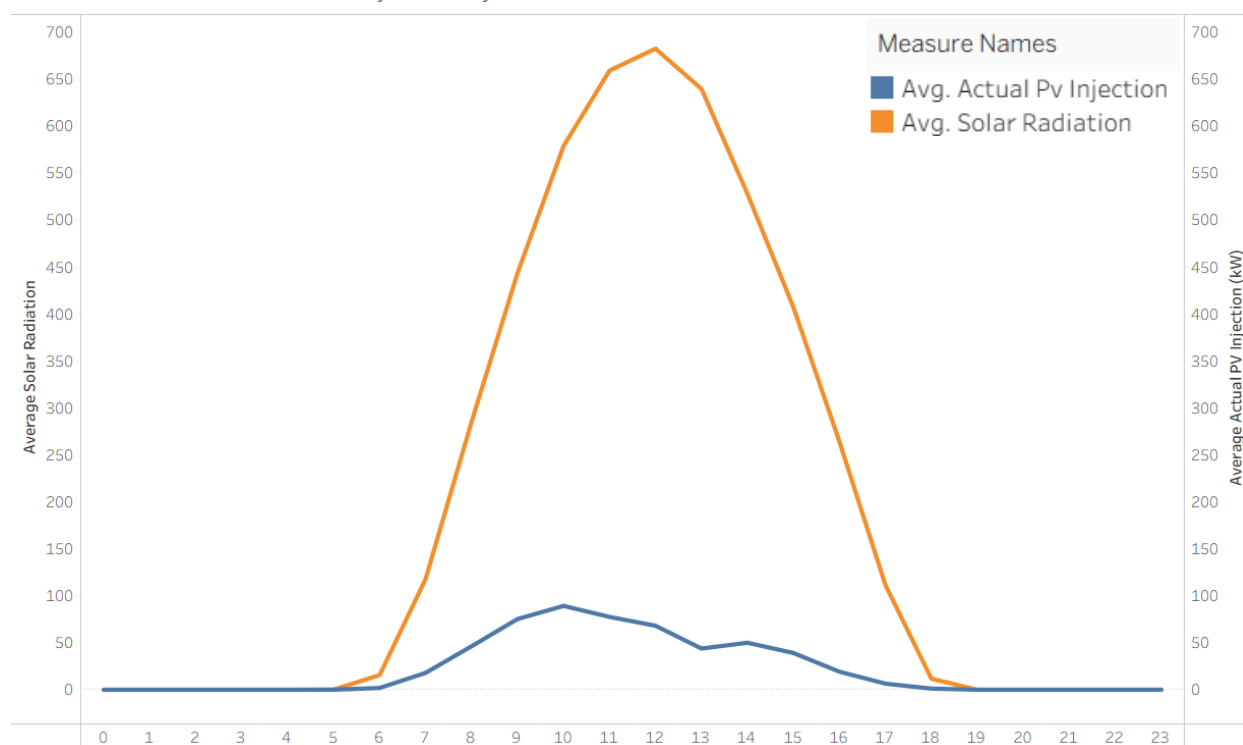


Figure 3. Comparison between average solar radiation and average actual photovoltaic injection throughout the day.

The visualization suggests the presence of operational constraints limiting photovoltaic energy utilization within the hybrid solar-diesel system. Even during periods of strongest solar resource availability, photovoltaic integration remained relatively restricted.

This finding directly supports the business question related to how solar radiation influences photovoltaic generation potential and how photovoltaic utilization varies throughout the day.

5.4.2 Unused PV Energy by Hour

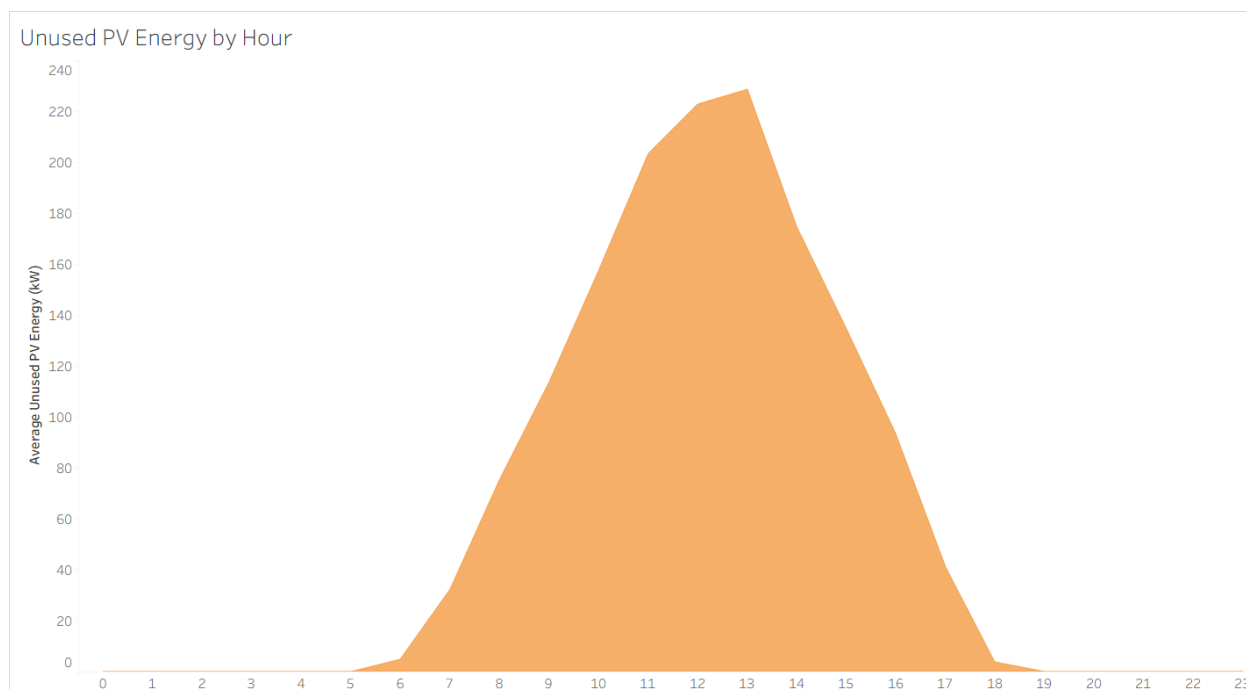


Figure 4. Average unused photovoltaic energy throughout the day.

The visualization reveals that unused photovoltaic energy increased considerably during peak solar production hours, especially between approximately 11:00 and 14:00.

The highest levels of unused photovoltaic energy coincided with periods of strongest solar radiation, reinforcing evidence of photovoltaic curtailment and limited renewable energy integration within the system.

This behavior indicates that a substantial portion of the available photovoltaic generation potential could not be fully injected into the electrical system during midday operational periods.

The visualization also suggests that operational optimization strategies or future energy storage solutions could potentially improve photovoltaic utilization and reduce renewable energy curtailment.

This finding directly addresses the business question related to how much photovoltaic energy remained unused due to operational constraints.

5.4.3 Monthly Actual PV Injection

Monthly Actual PV Injection

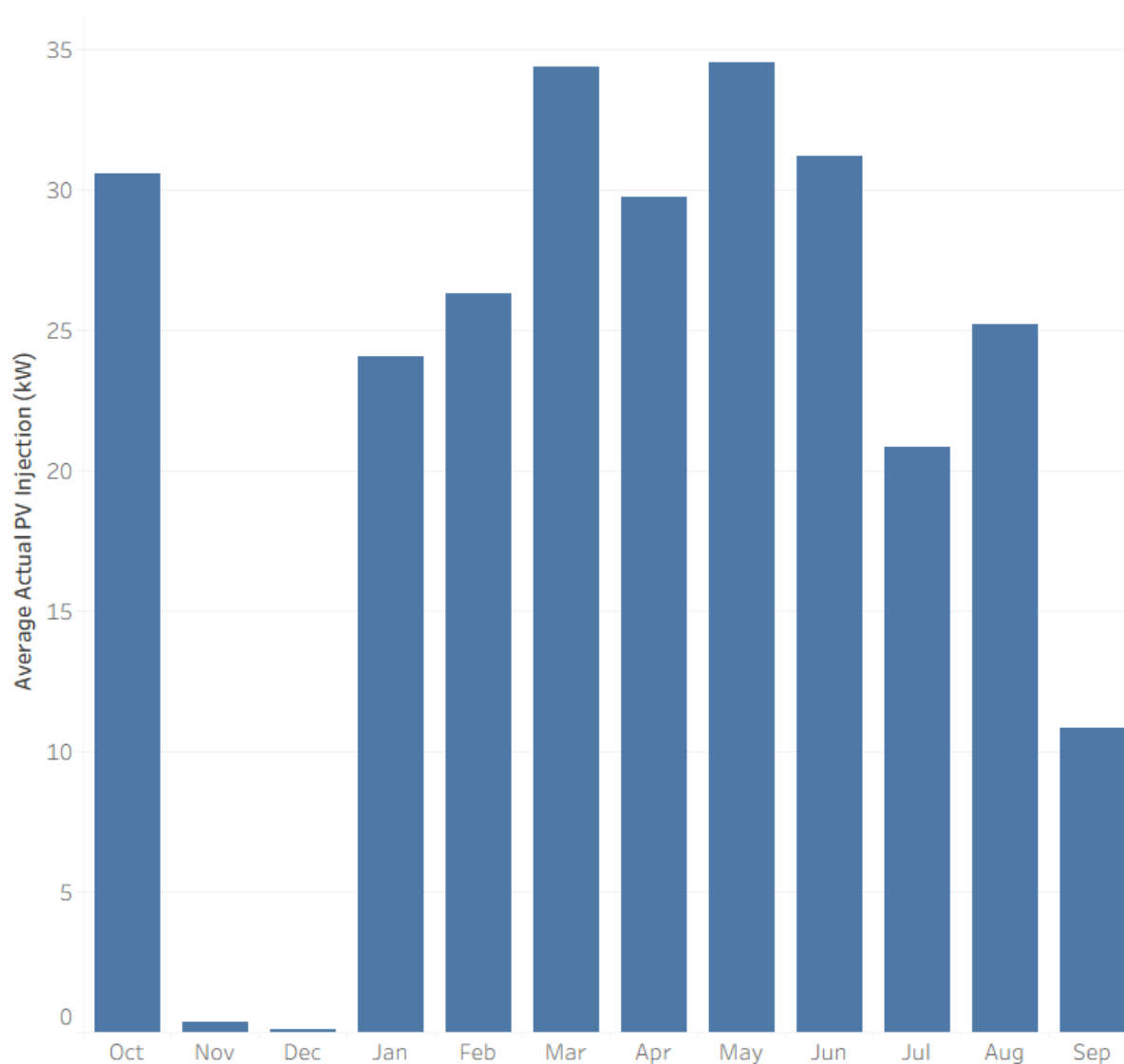


Figure 5. Average actual photovoltaic injection across analyzed operational periods.

The visualization shows considerable variations in photovoltaic energy utilization across the analyzed operational periods.

Higher average photovoltaic injection levels were observed during March and May 2019, while extremely low photovoltaic injection values were recorded during the selected operational periods from November and December 2018.

These findings suggest that photovoltaic utilization varied significantly between operational periods and that favorable solar resource availability alone did not necessarily guarantee high photovoltaic integration.

When compared with previous visualizations related to diesel generation and unused photovoltaic energy, the results reinforce evidence of operational limitations affecting renewable energy utilization within the hybrid solar-diesel system.

Because the dataset includes selected operational days rather than continuous monthly observations, the results should be interpreted as representative operational behavior for the analyzed periods.

This visualization directly supports the business question related to how photovoltaic utilization varied across different operational periods.

5.4.4 Monthly Unused PV Energy

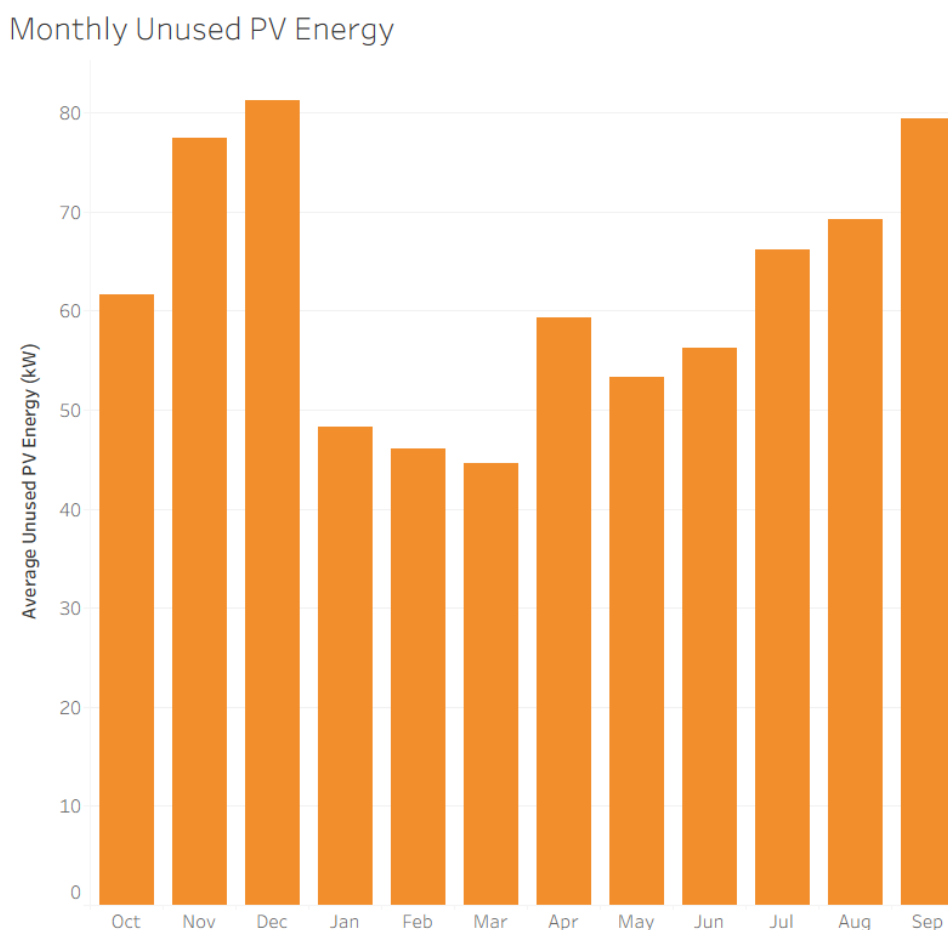


Figure 6. Average unused photovoltaic energy across analyzed operational periods.

The visualization shows noticeable variations in unused photovoltaic energy across the analyzed operational periods.

Higher average unused photovoltaic energy levels were observed during the selected operational periods from November and December 2018, as well as during September 2019.

These findings suggest that photovoltaic curtailment was not constant throughout the analyzed operational periods and may have been influenced by changing operational conditions, system demand behavior, or operational constraints affecting photovoltaic energy integration.

Despite the presence of available solar resource conditions identified in previous visualizations, substantial levels of unused photovoltaic energy remained during several operational periods, reinforcing evidence of limited photovoltaic utilization within the hybrid solar-diesel system.

Because the dataset contains selected operational days rather than continuous monthly observations, the visualization should be interpreted as representative of the analyzed operational periods rather than complete monthly operational behavior.

This visualization directly supports the business question related to identifying periods when solar energy was most underutilized.

5.4.5 Monthly Diesel Generation vs PV Injection

Monthly Diesel Generation vs PV Injection

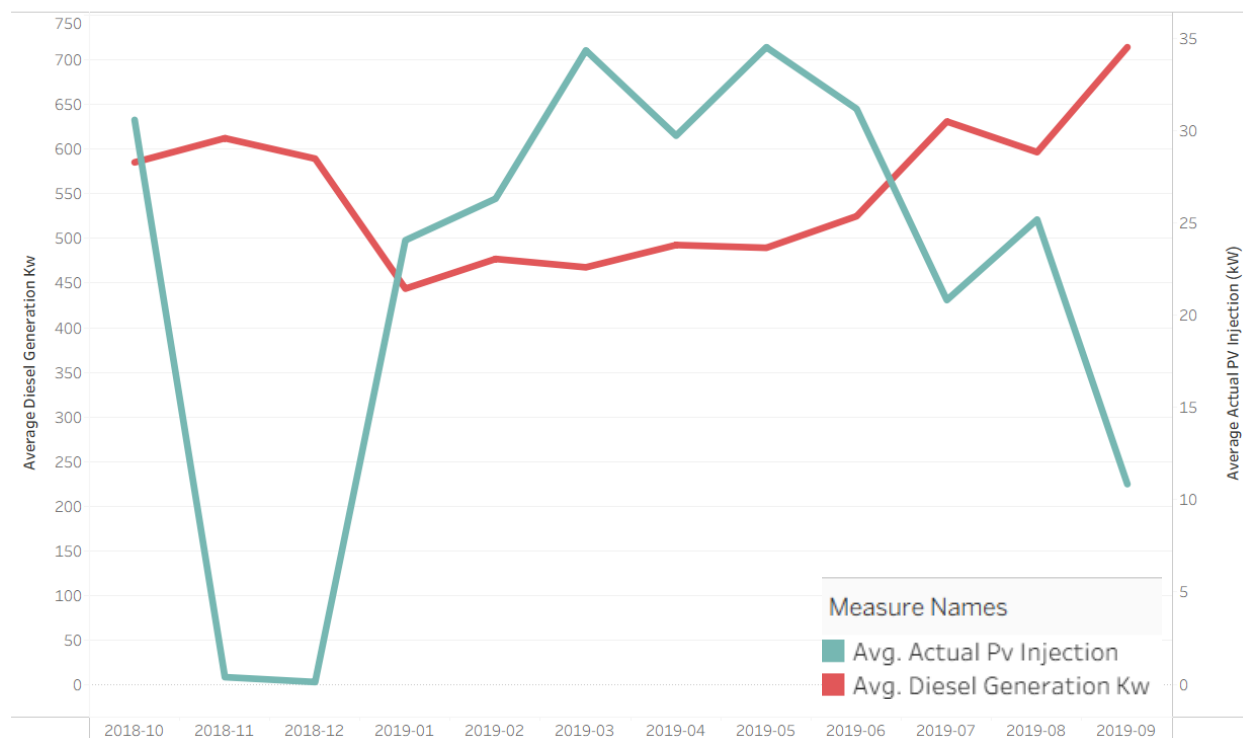


Figure 7. Comparison between average diesel generation and average actual photovoltaic injection across analyzed operational periods.

The visualization reveals that diesel generation remained consistently high throughout the analyzed operational periods, while actual photovoltaic injection remained comparatively limited and highly variable between operational periods.

Particularly important differences were observed during the selected operational periods from November and December 2018, when actual photovoltaic injection remained extremely low while diesel generation continued operating at relatively high levels.

Although the dataset contains selected operational days rather than continuous monthly observations, the analyzed records suggest that the hybrid system maintained considerable dependence on diesel generation even during periods with available photovoltaic generation potential.

When compared with previous visualizations related to solar radiation and unused photovoltaic energy, these findings may indicate the presence of operational limitations affecting renewable energy integration within the system.

This visualization directly supports the business question related to identifying periods of higher diesel dependence within the hybrid solar-diesel system.

5.4.6 Electrical Demand by Hour

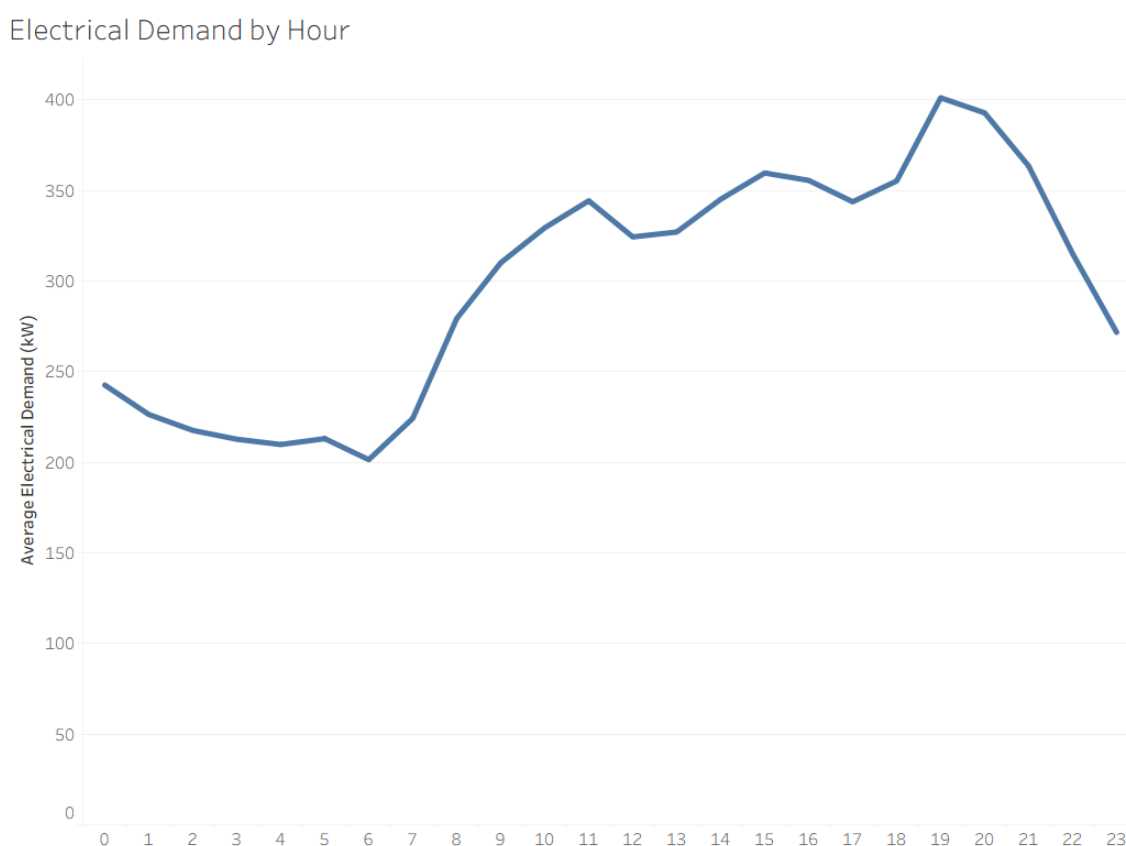


Figure 8. Average electrical demand by hour across the analyzed operational periods.

The visualization shows clear daily electricity demand patterns throughout the operational period analyzed.

Electrical demand remained relatively low during early morning hours and gradually increased during daytime and evening operation.

The highest average electrical demand levels were observed between approximately 19:00 and 20:00, when demand exceeded 390 kW. This behavior suggests that peak electricity consumption occurred primarily during nighttime operational periods.

When compared with previous visualizations related to photovoltaic injection and solar radiation, the results indicate that peak electricity demand did not fully coincide with periods of highest solar resource availability.

This operational behavior may partially explain the continued dependence on diesel generation during evening hours within the hybrid solar-diesel system.

The visualization directly supports the business question related to the relationship between electrical demand and photovoltaic energy injection.

5.5 Accessibility and Visualization Design Considerations

Several visualization design principles were applied during the Share phase to improve clarity, accessibility, and interpretability.

The visualizations were designed using:

- Consistent chart structures,
- Readable axis labels,
- Descriptive titles,
- Moderate contrast,
- Simplified layouts to avoid unnecessary visual complexity.

Dual-axis visualizations were used carefully in situations where comparing operational variables with different measurement scales improved interpretability. Colors were selected to help distinguish operational variables while maintaining readability.

Additionally, the visualizations prioritized operational meaning and stakeholder understanding rather than excessive technical complexity.

5.6 Key Findings Summary

The Share phase visualizations revealed several important operational patterns within the El Sena hybrid solar-diesel system:

- Solar radiation availability remained considerable during daytime operation, especially around midday periods.

- Actual photovoltaic injections remain significantly lower than the theoretical photovoltaic generation potential.
- Large amounts of photovoltaic energy remained unused during periods of strongest solar resource availability, indicating substantial photovoltaic curtailment.
- Diesel generation remained consistently high throughout the analyzed operational periods, suggesting persistent dependence on diesel-based electricity generation.
- Electrical demand peaked primarily during evening hours, when solar resource availability was limited.
- Photovoltaic utilization varied considerably across different operational periods, particularly during the selected operational periods from November and December 2018.

Overall, the visualizations reinforced evidence that operational limitations significantly affect renewable energy integration within the hybrid solar-diesel system, resulting in reduced photovoltaic utilization and continued dependence on diesel generation.

6. ACT PHASE

6.1 Purpose of the Act Phase

The purpose of the Act phase is to transform the analytical findings of the project into practical insights and potential operational recommendations. Based on the results obtained during the analysis and visualization stages, this phase focuses on identifying how stakeholders could use the findings to support decision-making related to renewable energy integration, photovoltaic utilization, and hybrid system operation.

This phase also explores possible opportunities to improve operational efficiency, reduce photovoltaic curtailment, and evaluate future strategies that could help increase the effective use of available solar energy within the hybrid solar-diesel system.

Additionally, the Act phase highlights how data analytics can support future operational planning, energy optimization initiatives, and further research related to renewable energy systems.

6.2 Final Conclusions

This project provided an operational analysis of the El Sena hybrid solar-diesel system using historical operational and environmental data collected between 2018 and 2019.

The analysis confirmed the presence of significant photovoltaic curtailment within the system. Although substantial solar resource availability was observed during daytime operation, only 26.8% of the theoretical photovoltaic generation potential was effectively injected into the electrical system, while approximately 73.2% remained unused.

At the same time, diesel generation remained consistently high throughout most analyzed operational periods, including hours with strong solar radiation availability. One of the most critical findings was the near-zero photovoltaic injection observed during November and December 2018 despite normal solar radiation conditions and without evidence of unusually high electrical demand. This behavior suggests the existence of operational restrictions or technical limitations affecting renewable energy integration.

The analysis also revealed a mismatch between peak solar generation and peak electricity demand. While photovoltaic generation potential peaked during midday periods, the highest electricity demand occurred during evening hours when solar radiation was unavailable. These findings highlight potential opportunities for operational optimization and future energy storage evaluation within the hybrid system.

Overall, the project demonstrated how data analytics techniques can support the evaluation of renewable energy utilization, operational efficiency, and decision-making processes in isolated hybrid energy systems.

6.3 Recommended Actions and Business Applications

6.3.1 Operational Optimization Opportunities

The analysis identified important opportunities to improve renewable energy utilization within the hybrid system.

Large amounts of photovoltaic energy remained unused during peak solar production periods, reaching average unused photovoltaic energy levels above 228 kW around 13:00. At the same time, diesel generation continued operating at relatively high levels despite favorable solar radiation conditions.

These operational patterns suggest that photovoltaic injection limits, dispatch strategies, or operational restrictions may be limiting renewable energy integration within the system. Reviewing photovoltaic dispatch prioritization and improving coordination between diesel and photovoltaic generation could potentially reduce diesel dependence and decrease photovoltaic curtailment.

Additionally, the mismatch between peak solar availability and peak evening electricity demand highlights the potential value of future battery energy storage solutions capable of shifting excess daytime photovoltaic generation toward nighttime consumption periods.

6.3.2 Renewable Energy Planning

The findings of this project could support future renewable energy planning and operational decision-making processes related to isolated hybrid energy systems.

The analysis demonstrated the importance of understanding the relationship between solar resource availability, electrical demand behavior, diesel generation, and photovoltaic integration

constraints. These insights may help stakeholders identify opportunities to improve operational efficiency and support future renewable energy expansion strategies.

Additionally, the analytical approach used in this project could be applied to similar hybrid systems operating in isolated regions where renewable energy integration remains operationally limited.

6.3.3 Monitoring and Data Analytics

Continuous operational monitoring and periodic analytical evaluation could help improve understanding of hybrid system behavior over time.

Future analytical initiatives could include:

- Larger operational datasets,
- Continuous daily operational records,
- Predictive analysis,
- Operational forecasting,
- More advanced visualization or monitoring systems.

The use of cloud-based analytical tools and data visualization platforms demonstrated how data analytics techniques can support operational evaluation and evidence-based decision-making within renewable energy systems.

6.3.4 Future Energy Storage Evaluation

Although large-scale battery storage systems may currently represent important economic and logistical challenges in isolated rural regions, future technological developments and cost reductions could eventually create additional renewable energy integration opportunities.

If economically feasible in the future, energy storage solutions could help reduce photovoltaic curtailment by storing excess solar generation during peak solar production periods and partially supporting electrical demand during evening hours when photovoltaic generation naturally decreases.

Future technical and economic feasibility studies could further evaluate the practicality of energy storage integration within hybrid solar-diesel systems operating in isolated regions.

The operational analysis also revealed a significant temporal mismatch between peak photovoltaic generation and peak electrical demand. While solar resource availability peaked during midday hours, the highest electricity demand levels occurred during evening periods when photovoltaic generation was naturally unavailable.

This behavior provides a strong technical justification for evaluating future battery energy storage systems capable of storing excess daytime photovoltaic generation and supplying part of the evening electrical demand. Such solutions could help reduce diesel dependence, decrease

photovoltaic curtailment, and improve renewable energy utilization within isolated hybrid systems.

6.4 Additional Data and Future Analysis Opportunities

Future studies could expand this analysis by incorporating continuous daily operational records, longer historical periods, fuel consumption data, operational maintenance records, and additional environmental variables.

The use of larger datasets could support more advanced analytical approaches such as predictive analysis, operational forecasting, anomaly detection, and renewable energy performance modeling.

Future analytical projects could also incorporate programming tools such as Python or R to support automated data processing, statistical analysis, machine learning applications, and more advanced operational simulations for hybrid renewable energy systems.

6.5 Final Reflection

This project demonstrated how data analytics techniques can be applied to evaluate operational behavior within hybrid renewable energy systems and support evidence-based interpretation of photovoltaic utilization patterns.

Using SQL analysis, cloud-based data processing, and Tableau visualizations, the project provided a structured analytical approach for identifying photovoltaic curtailment behavior, diesel generation dependence, and renewable energy integration challenges within the El Sena hybrid solar-diesel system.

Overall, the project highlighted the value of data analytics for supporting operational understanding, renewable energy evaluation, and future decision-making processes in isolated hybrid energy systems.